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Current practice in biodiversity impact assessment and prospects for developing an integrated process

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Irish guidance for Integrated Biodiversity Impact Assessment provides a methodological approach for integrating impact assessment requirements, with regard to biodiversity, under EU and Irish legislation. Preparation of the guidance was supported by extensive consultation, including international and national surveys. These offered insights into the issues affecting the treatment of biodiversity in impact assessment practice as well as expert opinion on factors affecting and supporting the development of a more integrated and adaptive approach. This paper contrasts the international literature with the consultation feedback. Among other aspects, the results highlight the requirement for an improved application of evidence-based assessment techniques, continuity in monitoring, enhanced information exchange between scientists, assessors and proponents, as well as increased awareness amongst stakeholders for inclusion of appropriate biodiversity protection objectives and mitigation in final planning decisions. Comparative analysis of results indicates that current practice is characterized by limited information exchange and the use of in-house databases in assessments. A central spatial data repository is identified as key for quantitatively assessing (cumulative) effects through Geographic Information Systems, and thus supporting evidence-informed decision-making towards biodiversity conservation.

Keywords: appropriate assessment; impact assessment; biodiversity impact assessment; biodiversity data; integration; international perspectives

Introduction

The need to track biodiversity responses to strategies for its conservation, or to predict, minimize or prevent its damage by development, is a research, practice and policy priority (Mörtberg et al. 2007; Slootweg et al. 2010; Wale & Yalaw 2010; Whelan & Fry 2011; King et al. 2012). Article 14 of the Convention on Biological Diversity (CBD) highlights the potential contribution that impact assessment can make to the conservation and sustainable use of biodiversity (Treweek 2001). Under the European Union (EU) legislative framework, impact assessment issues relevant to biodiversity arise with three key directives: the Habitats (CEC 1992), Strategic Environmental Assessment (SEA – CEC 2001) and environmental impact assessment (EIA; as amended, CEC 2011). The Habitats Directive subsumes assessment responsibility for the Birds Directive (CEC 1979, 2009) under the umbrella of Natura 2000 sites, while other provisions, notably the Water Framework Directive (CEC 2000) and the Environmental Liability Directive (CEC 2004), are add-ons to those directives as far as biodiversity assessment issues go. The requirements of these directives have been accordingly transposed into Irish legislation (e.g. European Communities (Environmental Assessment of Certain Plans and Programmes) Regulations, DEHLG 2004; European Communities (Birds and Natural Habitats) Regulations, DAHG 2011). Despite the overall conservationist approach of EU and Irish law, the varied requirements for addressing biodiversity within impact assessment requirements of these legislative instruments have

implications for the varying aspects of biodiversity, as per the official definition in the CBD. While the Habitats Directive focuses on the conservation status of habitats and species, SEA and EIA provide the opportunity to address wider biodiversity issues such as non-protected habitats, landscape connectivity or ecosystem services. Ideally, if – as in the EU – impact assessment of likely adverse effects on biodiversity (biodiversity impact assessment hereinafter, given the common use of the term in both Ireland and the UK) is required under different environmental legislation, the aspects of biodiversity covered under the different legislative instruments should be viewed as complementary. Therefore, the required processes should be integrated such that knowledge amalgamated under each process can be combined for a holistic assessment of potential effects on biodiversity (González et al. 2012). In this context, Appropriate Assessment (AA), a requirement to assess the potential for significant impacts on the conservation status of protected Natura 2000 sites under Article 6 of the Habitats Directive (CEC 1992), can be coupled with SEA and EIA (i.e. flora, fauna and biodiversity-relevant environmental factors) to ensure a more holistic approach to biodiversity impact assessment procedures that results in the conservation and sustainable use of biodiversity resources.

In view of the potential improvements that integration can bring to impact assessment of likely effects on biodiversity, the Irish Environmental Protection Agency commissioned research to develop a national procedure for *Integrated Biodiversity Impact Assessment* (IBIA).

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This led to the publication of Best Practice Guidance (González et al. 2012) and a Practitioner's Manual (González 2013). IBIA can be defined as a practical and systematic methodological framework that integrates SEA requirements with AA for plans, and EIA with AA for projects (González et al. 2012) (Figure 1). Where relevant, IBIA also addresses biodiversity-relevant requirements under the Water Framework, Environmental Liability and Floods Risk Directives (CEC 2000, 2004, 2007a). Despite observed variation in scope (e.g. legal implications and geographical extent) and practical approaches (e.g. extent of biodiversity considerations and habitat and species covered, scale and assessment detail or impact assessment techniques applied), the objective of biodiversity conservation is similar across the different appraisals (González et al. 2012). In light of this, the IBIA framework amalgamates assessment processes to enhance the congruence and efficiency of legal, administrative and operational processes. Coordination of assessments and planning processes was laid out to optimize time and resources and reduce/avoid duplication of efforts by improving communication channels and data sharing. Overall, the IBIA approach has been developed to enhance biodiversity impact assessment practice through integration of impact assessment procedures with regard to biodiversity, coordination and integration of specialist inputs, data, information and findings, and coordination of legal and administrative processes and decisions.

The resulting IBIA Guidance and Practitioner's Manual (González et al. 2012; González 2013) provide step-by-step recommendations on data gathering and management, communication and consultation, assessment methods and other relevant technical and practical considerations. The objective of this paper is to summarize and discuss the insights gained in the extensive international and national consultation on biodiversity impact assessment undertaken during the design phase of the IBIA, which subsequently contributed to such recommendations.

IBIA consultation and survey approach

National and international surveys were conducted in support of developing the IBIA framework. The surveys aimed to provide insights into current national and international biodiversity impact assessment practice issues. The overall objective was to explore critical aspects that facilitate or hinder integrated and adaptive biodiversity impact assessment across the world. The

questions aimed at capturing aspects of current practice to enable comparison between the Republic of Ireland (Ireland hereinafter) and other countries, as well as at gathering best practice recommendations (Table 1). For all the questions, multiple selection of answers was encouraged (with a defined set of options, including an open answer option), which is reflected in the overall percentage values presented and discussed below.

To exploit the potential for capacity-building based on shared international experiences and perceptions, it was considered vital to undertake international consultation. Therefore, a workshop was held at the 31st annual conference of the International Association for Impact Assessment (IAIA). It generated good debate on biodiversity impact assessment and provided the opportunity to distribute the questionnaire. As a follow-up, the questionnaire was circulated online to 132 IAIA members. Survey individuals were selected from the conference participants list to obtain an overview of professional and academic perceptions and opinions. Selection was carried out by applying the following criteria:

- As the project focused on biodiversity impact assessment (and European AA, SEA and EIA legislation and practice), the main selection criterion was participants' expertise. Individuals that lacked obvious biodiversity, ecology or impact assessment background were scoped out of the selection process.
- All European countries represented at the conference that fulfilled the above criteria were included in the emailing list. Where possible, two members from each country were included to optimize the geographic response.
- Worldwide, there is a variety of approaches, views and experiences in biodiversity impact assessment; therefore, at least one member with a biodiversity/ecological background from each non-European country represented at the conference was included in the email list.

Given the national context of the guidelines being prepared, an Irish workshop and survey were also carried out. The workshop was attended by 34 key governmental and consultancy representatives with SEA and/or AA expertise, and the resulting summary report (González et al. 2011b) was emailed to all attendees along with the questionnaire. The questionnaire was also circulated among additional 20 national experts to increase the survey pool and include other relevant expertise.

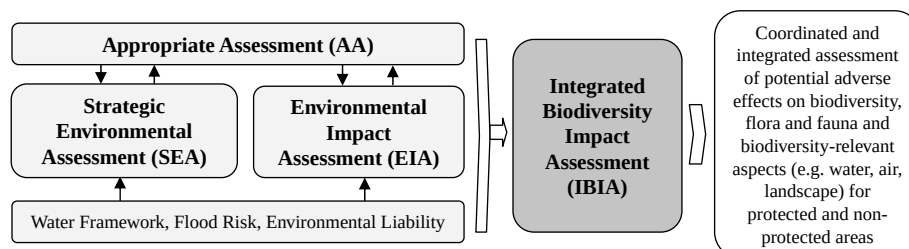


Figure 1. Simplified representation of the IBIA framework.

Table 1. Survey questions capturing current practice and best practice recommendations for biodiversity impact assessment

Current practice	
<i>Process coordination, communication and review</i>	
Q1	Does your country/institution/organisation coordinate biodiversity impact assessment to comply with different legal requirements such as those of SEA, EIA and AA?
Q2	Who are the main readers of biodiversity impact assessment reports?
Q3	Are biodiversity impact assessment reports in your country subject to quality control?
<i>Biodiversity data use and management</i>	
Q4	Which spatial datasets are most applicable in biodiversity impact assessment?
Q5	What other biodiversity relevant datasets do you have access to and apply in your country?
Q6	How are data gathered and managed to enable assessment of 'in combination' or cumulative effects of plans and projects (i.e. proposals)?
Q7	What do you consider to be the most common constraints to biodiversity data access and use?
Q8	What would you recommend to improve biodiversity data access and use?
<i>Biodiversity impact assessment approaches</i>	
Q9	What criteria are used to identify and assess 'ex-situ' effects?
Q10	Which is the most common biodiversity assessment technique applied by you/your country/institution/organisation?
Q11	What are the main constraints to applying GIS in biodiversity impact assessment?
Q12	Is there any guidance in your country with regards to biodiversity impact assessment techniques and methods?
<i>Scale and uncertainty</i>	
Q13	Uncertainty in biodiversity impact assessment is commonly associated with?
Q14	How is 'uncertainty' in biodiversity impact assessment results dealt with?
Q15	What are your recommendations to better address scale issues in biodiversity impact assessment?
<i>Potential to enhance current practice</i>	
Q16	Biodiversity impact assessment would be more effective for biodiversity conservation if:
	• Different impact assessments were combined into a central database • Long-term data of the vulnerable taxa were available • Your understanding of ecological processes were increased • Societal preferences with regard to biodiversity were more openly discussed • Administrative procedures of the biodiversity impact assessment process were simplified • Legal requirements to protect biodiversity were stricter • Opportunities for biodiversity enhancement were more commonly recognised • Biodiversity conservation were coupled with ecosystem services • Your country had more biodiversity experts • Biodiversity was monitored more closely

Results and analysis: international and national perspectives

Survey responses

The geographic distribution of the responses is presented in Table 2. Of 60 questionnaires distributed at the IAIA conference, 22 were returned at the end of the workshop

Table 2. Number of responses to the questionnaires and countries represented

Continent	Countries represented	Responses
Africa	Mozambique, Republic of South Africa, Uganda	5
Asia	India, Pakistan, Republic of Korea, Russian Federation	10
Oceania	Australia, New Zealand	3
Europe	Belgium, Czech Republic, Estonia, Italy, Netherlands, Spain, Sweden, United Kingdom	13
	Ireland	8
North America	Canada, Mexico, Trinidad and Tobago, USA	18
South America	Ecuador, Suriname	4
Total	24 countries represented	61

(36% response rate), along with 31 responses from 132 circulated online (23.5% response). In contrast, the percentage of national responses was lower, with eight of the 54 national questionnaires circulated returned (14% response rate). Despite the contrasting number of international and national responses, this paper aims at comparing international and national perceptions with the caveat that resulting percentages may not be directly contrastable. Nevertheless, the results enable contrast between national and international perceptions, as well as identification of international insights that can be applicable to enhance current practice in Ireland. The relatively high number of international responses (53 or 86.9%) provides valuable worldwide insights into current practice and best practice recommendations. However, they affect the overall results and the ensuing reflections are cautiously examined for their transferability to the national level as impact assessment requirements and procedures vary across the globe.

Current practice in impact assessment

Questions 1–8 (Table 1) aided in understanding current practice in the application of biodiversity impact assessment. These covered the coordination of processes, reporting and management of biodiversity datasets.

Coordination of biodiversity impact assessment across different legislative requirements

A large minority of both international (45%) and national (44%) respondents indicated that biodiversity impact assessment is commonly coordinated to comply with the different legal requirements under AA, SEA or EIA (Figure 2). This coordination seems more common at project level (37% international/22% national responses) than at plan/programme level (16 and 22%, respectively). This may be due to the more complementary scale and level of assessment detail required at EIA and project AA, and the potential contrasts in assessment detail between SEA and plan/programme AA. Given the worldwide responses, these findings may also suggest a lack of SEA

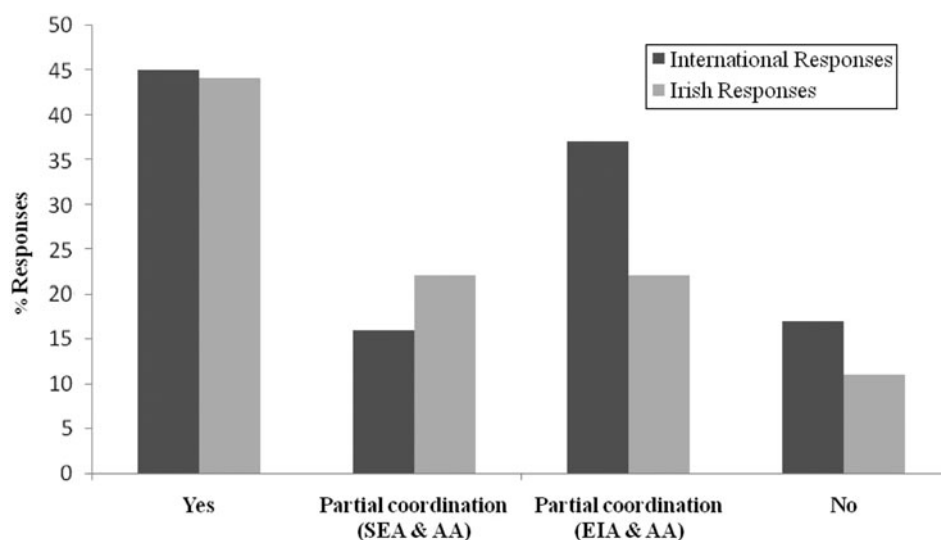


Figure 2. Questionnaire results showing the percentage of responses that indicated coordination, partial coordination or lack thereof between different EU legal requirements for impact assessment with regard to biodiversity aspects.

implementation in some countries. In all cases, it has been noted that consideration of biodiversity issues is poor, owing to the fragmented assessment of potential impacts a proposal may have (Byron & Treweek 2005; Gontier et al. 2006; Söderman 2009). Arguably, a tendency exists for parallel processes to develop in the EU whenever assessments are being conducted under the Habitats and SEA/EIA Directives. Such processes tend to involve several agencies with biodiversity interests, and result in isolated decisions being made without reference to the broader cumulative impact a proposal may have (Byron & Treweek 2005; Gontier et al. 2006; Söderman 2009; Fry et al. 2011a).

Quality control and communication during the assessment process

The majority (64%) of all the respondents indicated that biodiversity impact assessment reports (e.g. Natura Impact Statements under the EU Habitats Directive, or flora and fauna sections of SEA Environmental Reports or Environmental Impact Statements) were subject to quality control. Although quality control mechanisms (e.g. internal review, external peer review, etc.) were not explored through the survey, Irish best practice incorporates authority review by the Environmental Protection Agency and the National Parks and Wildlife Service for SEA Environmental Reports and Natura Impact Statements. Nevertheless, international and national studies have shown arrays of shortcomings in the quality of such assessments in the past (Byron et al. 2000; Rajvanshi 2001; Kolhoff & Sloodweg 2005; Gontier et al. 2006; Söderman 2009; Fry & O'Connell 2011). This highlights that evaluation criteria for reports may not reflect the requirements for effective biodiversity assessment (Fry & Scott 2011). Quality control should enhance communication, and its results need to be fed back into guidance on assessment processes.

Both internationally (34%) and nationally (50%), it was seen that planners/decision-makers are the main

readers of biodiversity impact assessment reports. Interestingly, plan/programme and project proponents were the least likely to read biodiversity assessments, which demonstrates a lack of communication of such potential impacts to relevant stakeholders (13% international/7% national). A shared understanding of the aims of biodiversity conservation and how impacts can be avoided through mitigation would require a common understanding by all professionals and stakeholders involved in or affected by biodiversity impact assessment practice (Treweek 2001; Hochstrasser 2013). Proactive communication between stakeholders and assessment teams early in the process contributes to impact assessment effectiveness (André et al. 2006).

Data use and management

As a result of regulatory assessments (e.g. AA, SEA and EIA), as well as research and project-based surveys, a broad infrastructure of primary biodiversity data is available. Nevertheless, it has been argued that existing primary biodiversity data are often inaccessible or undiscoverable (Chavan & Ingwersen 2009). When asked about accessibility to biodiversity-specific datasets, most respondents cited flora and fauna surveys (36% of international and 29% of national responses) with some respondents pointing out that any kind of species list or spatial classification of areas can be used in assessments. Invasive species surveys were more highly regarded nationally (25%) than internationally (16%). This may relate to Ireland's island status, but also to the Irish National Biodiversity Data Centre and the Invasive Species Ireland repository, which have positively contributed to an increased use of these datasets in relevant assessment. In comparison with other spatial datasets, site surveys (18% international/14% national) and habitat maps (18 and 17%, respectively) were considered the most applicable to biodiversity impact assessment (Figure 3). It has been acknowledged that high-quality primary

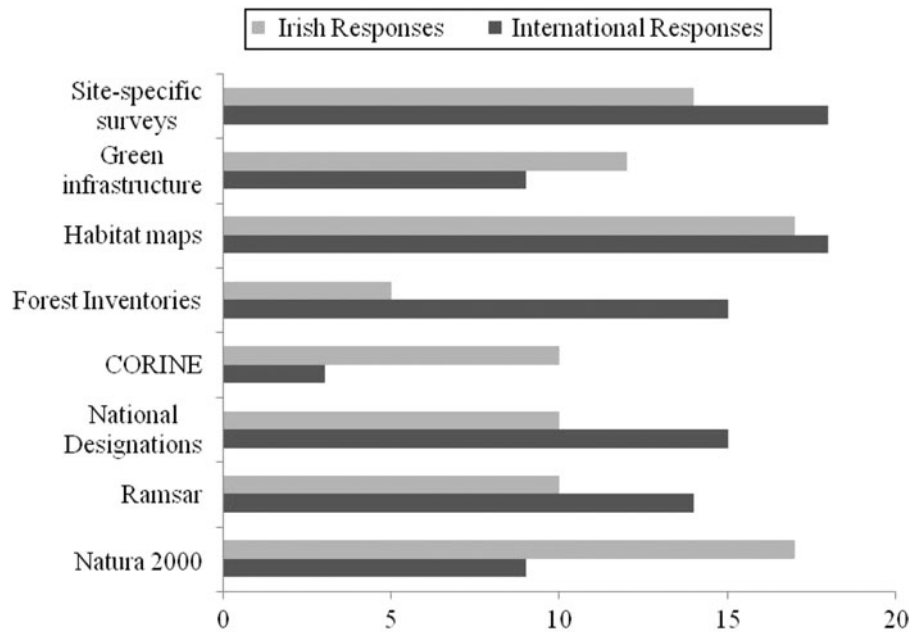


Figure 3. Questionnaire results showing the spatial datasets most applicable in biodiversity impact assessment.

biodiversity information gathered from site-specific surveys can aid in determining potential impacts on biodiversity more readily than secondary information (e.g. water quality, soil classifications, etc.), as they provide ground-truthed biodiversity evidence (Gontier et al. 2006; Wale & Yalaw 2010; González et al. 2012). Thus, flora, fauna, invasive species surveys and habitat maps are seen as most applicable and available worldwide for providing the evidence base (i.e. facts and findings) in both survey responses and international literature (González et al. 2012; King et al. 2012; Noble et al. 2012).

Only 24.5% of international survey participants worked in the EU, which led to clear differences in the use of European datasets such as Natura 2000 and the European CORINE landcover classification. Natura 2000 datasets were attributed a higher importance in Ireland (17%) than internationally (9%), whereas the use of national designations seems more widespread internationally (15%) than in Ireland (10%). CORINE datasets were considered least applicable in biodiversity impact assessment (3% international versus 10% national). Despite Irish respondents referring to CORINE, it should be noted that this is a proxy dataset and its applicability in biodiversity studies is therefore rather limited. A 5ha unit is currently available (i.e. CORINE 1990–2006) for identifying landcover changes over time, but the planned 1ha refinement will increase its applicability across the EU, particularly in plan-level AAs and SEAs. This will also benefit location of the qualifying interests and habitat boundaries within sites, which can be difficult to determine from existing datasets (e.g. Natura 2000). Other relevant datasets included green infrastructure and ecological corridors (9% international and 12% national responses) and forest inventories (15% international/5% national). Overall, it seems that, while datasets that are specifically aimed at describing biodiversity, such as site-

surveys, habitat maps and Natura 2000 and other national designations, are favoured, the potential of other supporting datasets is generally recognized, but not embraced with the same frequency. This may partially be due to a lack of knowledge amongst practitioners about other biodiversity-related data, or limitations as to how to incorporate such related information into assessments.

In all cases, and as supported by international literature (e.g. Chavan & Ingwersen 2009; González 2012), various residual barriers impede data access. These can be technical infrastructural obstacles (such as the lack of a national biodiversity data repository) and/or data quality issues (e.g. precision, quality, accuracy and scale), making datasets created as non-reusable ‘one-offs’. Intellectual property rights can also limit the type and amount of information published and its accessibility for future assessments (Gontier 2007; Söderman 2009; Wale & Yalaw 2010; King et al. 2012). Access to, and use of, biodiversity datasets is constrained (Figure 4) by lack of knowledge of their availability (24% international/21% national responses) and, with contrasting international/national emphases, by scale and copyright/licensing (25% and 8% of international, vs 6 and 16% of national responses, respectively). Identifying metadata problems as a common constraint to data access and use scored higher nationally (14%) than internationally (7%) – possibly because of the recent wider availability of metadata in Ireland as a result of the implementation of the INSPIRE Directive (CEC 2007b). Metadata are particularly important when re-using data gathered during previous assessments, as survey methods and level of detail tailored to differing study requirements must be taken into consideration when establishing their fitness for use and purpose (Söderman 2009; González 2012). Internationally, quality issues seemed more problematic (21%) than nationally (10%). The lack of knowledge about available

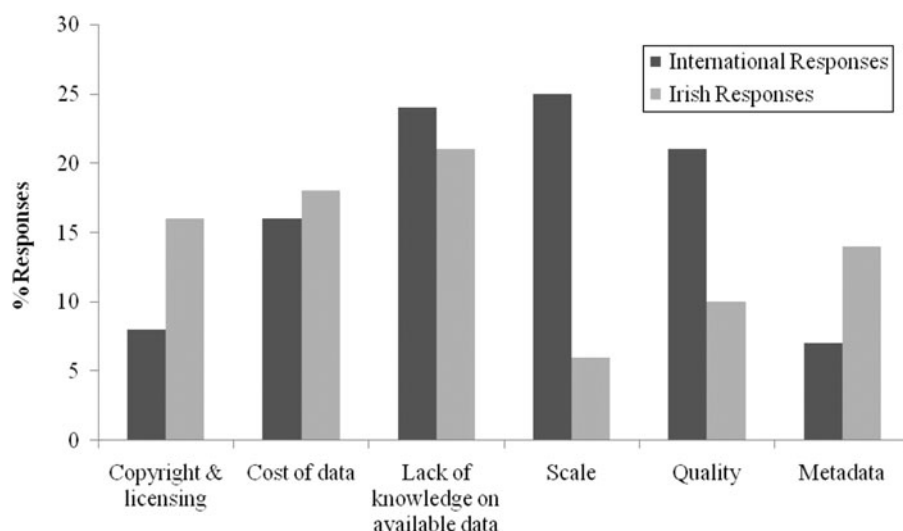


Figure 4. Expert views on the most common constraints to biodiversity data access and use.

datasets noted by respondents results in a weak information basis, leading assessments to generally focus on local protected species and sites (Gontier et al. 2006; Söderman 2009), rarely considering the ecosystem level and commonly omitting unprotected species and areas of biodiversity value that may fulfil important ecosystem services. The effects of data gaps and inefficiencies related to accuracy, completeness and scale (among others) in impact assessment have been widely reported (e.g. Byron & Treweek 2005; Söderman 2009; González 2010; González et al. 2011a, 2011b).

Gathering and managing data to enable the assessment of in-combination or cumulative effects of proposals was most commonly done through in-house databases from initial or previously approved proposals (36% international/28% national responses). This alludes to a lack of readily available external sources of biodiversity information, or poor collaboration between agencies that could aid in identifying potential impacts. Difficulty in accessing biodiversity datasets can render valuable information for determining cumulative or in-combination effects inaccessible, therefore impeding fully evidence-based assessments, and this commonly occurs where (as in Ireland) no central repository exists (Söderman 2009; González et al. 2012; King et al. 2012). Internationally, there is no indication of multiple stakeholder collaboration being used in data gathering and creation (Canter & Ross 2010; King et al. 2012). A noteworthy minority of respondents stated that they had no data management standards to facilitate a comprehensive prediction of impacts on biodiversity, more so internationally (23%) than nationally (10%). Without proper facilities to gather and manage data, or any mechanism to share information between assessments, long-term improvement in understanding impacts is rendered impossible – as is the proper identification of in-combination or cumulative impacts. Shortcomings in assessment of cumulative and in-combination effects have been reported internationally, noting a weak information basis and lack of proper regard to ecological structures as key constraints, together with a lack of commitment by

decision-makers to ‘good practice’ (Thompson et al. 1997; Byron et al. 2000; Söderman 2005; Geneletti 2006; Mörtberg et al. 2007; Söderman 2009; Canter & Ross 2010).

Techniques for impact assessment, treatment of scale and uncertainty

Questions were asked regarding techniques used in impact assessment and how scale and uncertainty are accordingly addressed (Table 1) – not only to describe current international practice but also to identify possibilities that are underused in Ireland. Common AA, SEA and EIA techniques include site-specific surveys, matrix-based expert judgement, ecological modelling and spatial assessment using GIS (Gontier et al. 2006; Wale & Yalaw 2010; Weingarten 2010).

Scale in impact assessment is at the discretion of the assessor, and issues can arise in determining what scale to apply (González et al. 2011a). A key recommendation by international and national respondents to better address this was the application of techniques appropriate to the scale of the study (39% of total responses). In line with this recommendation was the remark of an Irish respondent who noted that some aspects of biodiversity impact assessment, in particular those under the Habitats Directive, were difficult to assess at the strategic planning level (e.g. county development plans). This reflects a conflict between the goals set by legislation and the level of detail required from an ecological point of view for an efficient and effective assessment. Applying multiple assessment scales was more highly regarded internationally (32%) than nationally (15%). The use of available datasets was also recommended internationally (29%) and nationally (31%), as long as scale and any other data limitations are clearly acknowledged. These findings reinforce other international reviews emphasizing the importance of choosing the correct scale(s) and reporting on any data limitations when evaluating potential impacts on biodiversity. The use of arbitrary scales and geographical extents that are not determined by ecological

characteristics or processes and that may rely only on available data can result in important ecosystem features and species unknowingly being screened out of the assessment (João 2002; Gontier 2007).

Survey responses suggested that the most commonly applied biodiversity assessment technique is site-specific assessment (Figure 5); this is perceived to be more common nationally (44%) than internationally (27%). GIS-based spatial assessment was also perceived as common, albeit less so nationally (14%) than internationally (27%). Although GIS methods have been extensively applied to impact assessment and their positive contribution to decision-making has been widely acknowledged (e.g. Aspinall & Pearson 2000; Münier et al. 2004; Mörtberg et al. 2007; Geneletti 2008), limitations in the development of appropriate GIS modelling techniques to efficiently assess complex environmental interactions may have aided this trend in responses, as practitioners are commonly unaware or rarely use such tools (Atkinson et al. 2008; Atkinson & Canter 2011b). This is further emphasized by the fact that the use of ecological modelling is deemed non-existent nationally, while only 8% of international respondents indicated using it. This shows a clear discrepancy between work in biodiversity impact assessment and conservation biology, where GIS and modelling are commonly used (Gontier et al. 2010; Fry et al. 2011b). In addition, limitations in the availability and accessibility of spatial data for impact assessment, together with time and resource constraints with regard to relevant skills, may further limit the applicability of GIS (González 2012). This was supported by the survey, as 32% of international and 63% of national respondents saw a lack of spatial data as the main constraint to GIS application in biodiversity assessment (followed by time and resource issues – 25 and 13% of responses, respectively). This is reinforced by international literature showing similar limitations to using GIS in impact assessment (Gontier et al. 2006; Atkinson & Canter 2011b; González et al. 2011a; González 2012; Noble et al. 2012). The results also indicated that matrix-based expert judgement is least commonly applied, both internationally

(7%) and nationally (13%). Such assessments can lack objectivity and do not enable quantification in impact assessment, limiting their applicability and making them useful only when spatially specific or quantifiable biodiversity datasets are unavailable (Therivel 2004; Atkinson & Canter 2011b; González et al. 2011a; Noble et al. 2012). Overall, the results suggested that evidence-based techniques that need expert input, such as ecological modelling and ecosystem-based approaches, are not used as frequently as the more traditional site-based assessment. This could potentially be addressed through institutional arrangements that allowed for tighter collaboration between scientists and practitioners (Fry et al. 2011b).

Examining assessment methods and approaches further, the results indicated that disturbance of adjacent habitats/lands provide the most common criteria used to identify and assess ex-situ effects, both internationally (66%) and nationally (42%), followed by potential risks to water-dependent habitats (34% internationally/31% nationally). This limited consideration of aquatic biodiversity in impact assessment, despite its importance, has recently been highlighted elsewhere (Grossman 2013). This has important implications for developing guidance for biodiversity assessment within the remit of, for example, the EU Water Framework Directive. It also suggests that ecologists conducting assessments may generally have a background in terrestrial ecology and are thus less inclined or able to include proper considerations for aquatic habitats.

A majority (55%) of international respondents indicated a lack of guidance on biodiversity impact assessment techniques and methods (although relevant international guidance is available, such as the CBD Guidelines 2006), whereas 71% of national respondents acknowledged having guidance (including National Roads Authority Guidelines, Heritage guidance on habitat surveys and Department of Environment, Community and Local Government guidance on implementing national legislation). No one technique is 'fit-for-all' or fully effective; all are associated with a certain level of uncertainty resulting from ecological data limitations (36% international/43% national responses), or insuffi-

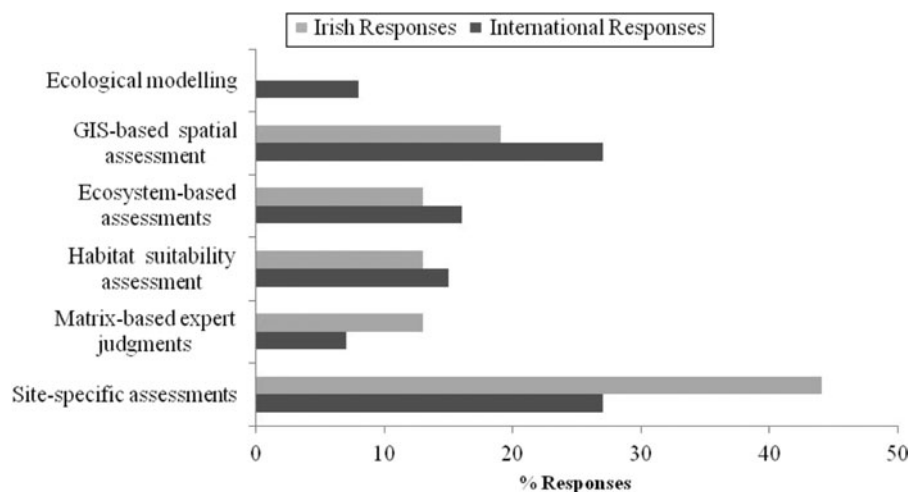


Figure 5. Expert views on the most common biodiversity assessment techniques applied.

cient data (23 and 22%, respectively). Interestingly, model and GIS parameterizations and assumptions were regarded as least associated with uncertainty in biodiversity impact assessment (12 and 7%, respectively).

Using existing datasets while acknowledging all data limitations was deemed good practice in addressing uncertainty (29% international/31% national), and is supported by international literature (Treweek 1996; Geneletti et al. 2003; Gontier et al. 2006; Partidário 2007; King et al. 2012). A majority of consulted practitioners and experts considered applying additional mitigation measures to be the best way of dealing with uncertainty (60% international/52% national). This clearly indicates that monitoring the effectiveness of mitigation would significantly contribute to understanding impacts and improving worldwide practice. Internationally, a flexible consent process was seen as another good practice to deal with uncertainty (38%), and that adding contingency and monitoring plans could aid in this, with a two-stage consent process being considered the least viable option (9% international/10% national), despite it presenting an opportunity for additional data gathering. An integrated approach is sought by the EU (González et al. 2012) as issues can arise when a two-stage consent process is in place, particularly with regard to fragmentation of the assessment that can make determination of in-combination and cumulative effects difficult and result in the continued placement of biodiversity issues on the periphery in the assessment of proposals.

Aspects contributing to adaptive biodiversity impact assessment

An integrated approach needs to be adaptive in order to accommodate changes in legislation and advances in methods/techniques and thus ensure its effectiveness and efficiency over time. An adaptive IBIA approach also promotes improved assessments through enhanced data

sharing and communication, thus leading to better knowledge. When considering the different options, it was clearly shown that practitioners are well aware of the values of long-term observational data as well as central data repositories for improving impact assessments (Figure 6). Respondents answered that, if long-term data on vulnerable species were made available, biodiversity assessments would be more effective (16% of international and 33% of national respondents). Such information would aid in the protection of sensitive species and habitats covered by the Natura 2000 network (CEC 1992) and red-data listed species (IUCN 2013). The development of national repositories for biodiversity data would allow the accumulation of long-term data and, therefore, contribute to determination of cumulative, synergistic and in-combination effects (Gontier et al. 2006; González et al. 2012; King et al. 2012). This is supported by survey findings indicating that, if separate impact assessment results were accumulated within a central (national or global) database, assessment effectiveness would be significantly improved (14 and 23% respectively). Such a data repository could also aid in developing an understanding of ecological processes, although a lack of such understanding was only remarked upon by international respondents – the small national sample size possibly contributing to this result. The incorporation of in-house datasets and their effective publication online would increase the availability of evidence-based information to practitioners and enable a more effective identification of potential (cumulative) effects, encouraged by the use of spatial data in planning and decision-making promoted by the INSPIRE Directive (CEC 2007b). In protecting biodiversity, the stricter enforcement of legal requirements was regarded as having more effect in an Irish context (15%) than in an international context (10%). National and international respondents reacted differently when judging the importance of activities interpretable as contributing to the development of a

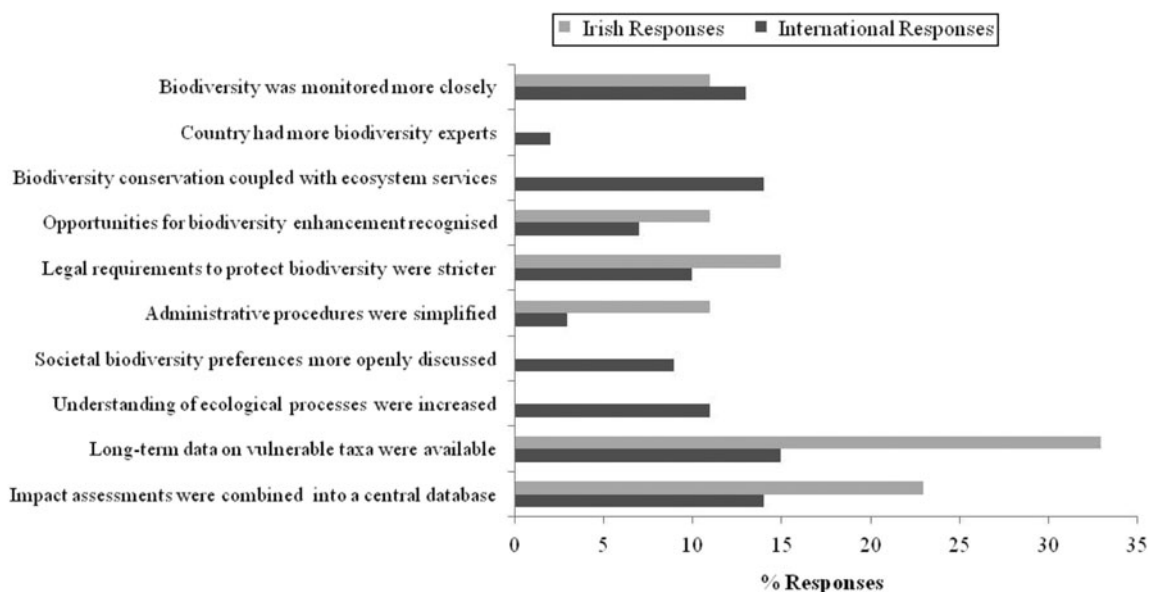


Figure 6. International and national expert opinion, on the factors that would contribute to more effective biodiversity impact assessment and conservation.

common framework in the assessment process. Whereas opportunities for biodiversity enhancement and conservation were more commonly recognized by national than international respondents (11%/7%), only international respondents (9%) thought it necessary to hold a discussion on societal preferences regarding biodiversity and ecosystem services. The contrasting results may reflect the fact that ecosystem services are rarely integrated into current Irish impact assessment practice. Interestingly, having more biodiversity experts was regarded as the least critical factor for improving impact assessment and biodiversity conservation (2% international/0% national).

Summary of findings: factors supporting integrated biodiversity impact assessment

The surveys gathered international expert opinion on critical aspects that facilitate or hinder effective and integrated biodiversity impact assessment. This section summarizes key survey findings and supporting international literature that have been incorporated into, or addressed by, the IBIA framework.

Articles 5 paragraphs (2) and (3) of the SEA Directive, while not being prescriptive, clearly encourage integrated, or at least coordinated (i.e. simultaneous and harmonized), assessment (CEC 2001). In particular, Article 11 paragraph (2) notes: *‘For plans and programmes for which the obligation to carry out assessments of the effects on the environment arises simultaneously from this Directive and other Community legislation, Member States may provide for coordination or joint procedures fulfilling the requirements of the relevant Community legislation in order, inter alia, to avoid duplication of assessment’*.

A large minority of national and international respondents indicated that biodiversity impact assessment techniques were being applied when carrying out AA, SEA and EIA, with the result of biodiversity considerations being provided for in assessment processes. This is seen to occur more at project level, as the ability to carry out site-specific surveys for identifying impacts on flora and fauna is more practical/feasible at that level. The survey findings indicate that planners/decision-makers are the main readers of biodiversity impact assessment reports, and that assessments are subject to quality control. The latter is encouraging as more quality information and detailed and comprehensive assessments are needed to comply with legislative requirements that enforce the protection of biodiversity (CEC 1992, 2001, 2009, 2011).

The survey included specific questions to address data management and applied assessment techniques as the resulting IBIA manual was to provide specific guidance on these aspects. Moreover, these considerations (e.g. data limitations and expert assessment tools and methods) significantly affect assessment results (João 2002; Geneletti 2006; Partidário 2007; González 2012). The results clearly indicated that biodiversity impact assessment practice is still heavily reliant on site-specific survey data produced by ecological consultants. The main obstacles for use of other data cited were lack of knowledge of datasets, as well as scale and copyright issues. Lack of knowledge can be

readily addressed by providing practitioners with a list of available datasets. Such a list has been developed as an appendix to the IBIA guidance (González et al. 2012). Although it was clearly recognized by respondents that using a scale appropriate to the scope of assessment was of prime importance, they still indicated that site-level information was the most useful. This reliance on site surveys may indicate that high-resolution localized information is needed for effective assessment, but may also show that respondents are less familiar with the potential of broader-scale datasets to inform assessment processes – particularly relevant at higher planning tiers. In light of this, guidance on the application of multiple assessment scales and techniques to combine localized information with broader-scale information was included in the IBIA framework. Moreover, the IBIA framework recommends anticipation and highlighting of potential shortcomings in terms of data scale and availability at higher planning tiers, so that the necessary data collation is undertaken for project level assessments.

Nevertheless, the responses also indicated that a wealth of information exists from in-house databases of approved and initial proposals; information that can aid in identifying cumulative and in-combination effects, since flora and fauna datasets (along with Natura 2000 sites, green infrastructure and ecological corridors, and forestry inventories) are seen as readily accessible and applicable to impact assessment. In Ireland, the increased external information available through the creation of the National Biodiversity Data Centre further assists IBIA, allowing practitioners access to more biodiversity-relevant datasets. The future for these datasets seems positive as site-specific assessments and surveys were perceived as a common biodiversity assessment technique worldwide and a prime data source. Such surveys will continue to be carried out, increasing our understanding of ecological processes and helping to address the ‘fortress’ conservation approach to impact assessment. The National Biodiversity Data Centre’s online repository could be further enhanced by incorporating AA/SEA/EIA outcomes relevant to biodiversity conservation, which could be then transferred into the Global Biodiversity Information Facility. This would strengthen the verifiability of impact assessment findings, reduce uncertainty, improve transparency of impact assessment results and support evidence-based decision-making processes, thus aiding the applicability of IBIA.

Throughout the survey, the need was highlighted for a centralized and standardized biodiversity database that gathers long-term datasets on vulnerable species and habitats. Such a database would help avoid duplication and facilitate (cumulative) effects assessment and monitoring. Respondents indicated that in-house databases from previous proposals are mainly used in assessing impacts, which suggests lack of collaboration between agencies and private practitioners owing to an absence of external data collating sources. Arguably, insufficient data are available to determine in-combination and cumulative effects, resulting in poor application of cumulative effects assessment, which both survey respondents and international literature indicate is an underused

technique (Baxter et al. 2001; Canter & Ross 2010; Atkinson & Canter 2011a). A centralized database would not only increase information exchange between practitioners, but could also be used by scientists to increase the background knowledge on general biodiversity management (Fry et al. 2011b). Furthermore, detailed metadata and data quality control mechanisms necessitated by centralized repositories would address the lack of data management techniques apparent from a significant proportion of responses. IBIA therefore highlights the maintenance of such a central data repository – favouring the National Biodiversity Data Centre for the Irish context (González et al. 2012).

The lower use of GIS in Irish assessments had been anticipated and was confirmed by the survey. As a result, extensive guidance on how to apply GIS was included in the IBIA methodology. GIS-based approaches enable case-by-case appraisal of key potential land-use conflicts (Atkinson & Canter 2011b; González et al. 2011a; González 2012), while composite maps aid in identifying potential cumulative effects (Hegmann et al. 1999; Cooper 2010; Atkinson & Canter 2011b; González et al. 2011a).

Given an open-access spatial repository, the use of GIS in Irish biodiversity impact assessment could be increasingly relied upon, reversing what national survey respondents indicated as being currently poor. GIS and spatial data can play a significant role in IBIA, as well as worldwide biodiversity impact assessment approaches, by providing spatial evidence, and integrating and objectively assessing multiple biodiversity-relevant datasets. Spatial assessments that promote quantitative approaches and address in-combination/cumulative effects, and their resulting maps, can convey baseline and assessment results in a concise and visual manner that improves understanding of all stakeholders on potential biodiversity impacts. The survey reveals encouraging signs for the use of GIS in impact predictions, albeit more so internationally than nationally. GIS tools can assist in rapidly bringing spatial data together, overlaying them with the proposal boundary and thus undertaking preliminary assessments of biodiversity and other aspects of the environment supporting evidence-based assessments (González et al. 2011a). GIS-based approaches can enable case-by-case screening for AA and SEA/EIA and facilitate the identification of key aspects during SEA/EIA.

Residual issues associated with GIS-based assessment are important. Not surprisingly, the lack or poor quality of ecological data was cited by both Irish and international respondents as the most significant source of uncertainty in impact assessment, a finding supported by international literature (Geneletti 2006; Partidário 2007; González 2012). Uncertainty is inherent in complex socio-ecological systems; the only way of addressing it is to adopt an adaptive framework that includes monitoring and an effective system for feeding information gathered back into the assessment process to thus promote ongoing improvement of the quality of assessments and findings (Mitchell 2009; Hochstrasser 2013). Whereas the majority of respondents cited additional mitigation measures as the common way of addressing uncertainty, the potential for

monitoring and flexibility in the consent procedures was also recognized. The importance of monitoring was highlighted in the IBIA guidance, which also included key relevant indicators to assist both assessments and monitoring of implementation.

There were interesting similarities and differences between Irish and international respondents. Both groups thought that assessment processes under different legislative requirements were at least partially integrated. This is surprising as disjointed assessment and consent processes under different legislation have caused significant delays in the Irish planning process. Indeed, the IBIA methodology was developed to improve communication and information exchange between biodiversity impact assessments under different legislative requirements, and will hopefully end this disjointed practice. Lack of communication with all stakeholders, including proponents, was also highlighted by both groups of respondents. Even though the proponent is not technically required to be aware of the assessment findings, these are vital for the design and incorporation of effective mitigation options, as well as for the identification of opportunities for biodiversity enhancement as part of the proposal. Whereas the importance of the latter was emphasized more by Irish respondents, discussion on priorities for biodiversity and ecosystem services was considered relevant by international respondents to improve biodiversity impact assessment practice. Ongoing dialogue with proponents, starting at the screening stage, is therefore essential for effective biodiversity assessment. The IBIA framework incorporated step-by-step recommendations on how to promote ongoing communication and ensure that biodiversity considerations are enshrined in all steps of a proposal from draft to completion-which are applicable, in particular, to the Irish planning system and transferable to other EU and worldwide jurisdictions.

Conclusions

Feedback gathered through international and national consultation suggested a general support for integrated assessment practices, indicating that this is potentially achievable worldwide, despite varying national legislative arrangements. A holistic approach to biodiversity impact assessment can contribute to enhancing the effectiveness of impact assessment and, as a result, to improving biodiversity protection for both designated and wider biodiversity areas. Nevertheless, survey results highlighted persistent issues with data quality, scale and uncertainty which, in turn, affect the applicability of spatially-specific and quantitative assessment techniques that also address cumulative effects. Therefore, integration of processes is not sufficient for improving international practice; baseline information needs to be improved worldwide and assessment techniques need to capture cumulative effects. In addition, the results highlight that lack of communication between assessment teams, stakeholders and proponents can impede efficient incorporation of biodiversity concerns into the proposal in order

to ensure biodiversity conservation. Ongoing communication, collaboration and dialogue have repeatedly been underlined as vital for effective impact assessment; tighter collaboration with specialized scientists may also help address persistent biodiversity knowledge issues.

Survey responses conveyed a number of recommendations that align with those identified in international literature, including the creation of a centralized repository of biodiversity information – already initiated in Ireland through the National Biodiversity Data Centre and worldwide through the Global Biodiversity Information Facility. Arguably, mandatory requirements for data upload and sharing would contribute to strengthening national and international biodiversity data-gathering initiatives, as well as to minimizing duplication and supporting assessment of cumulative effects. In order to improve the fitness for purpose of these data repositories for the specific task of biodiversity impact assessment, they should be directly and systematically linked into effective monitoring programmes.

Although it is widely acknowledged that further efforts are needed to improve current AA, SEA and EIA practice, amalgamating assessment processes and coordinating data gathering, assessment and communication through an integrated process such as the IBIA can significantly contribute to providing a more holistic approach to biodiversity (and environmental) assessment.

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